

CS & IT ENGINEERING

Digital Logic

Sequential Circuits

Lecture No 03



By- Aditya sir

Recap of Previous Lecture



Topic



Intro to Sequential Circuits



Latches

Topics to be Covered



Topic



Latches

S-R FlipFlop



About Aditya Jain sir



1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.

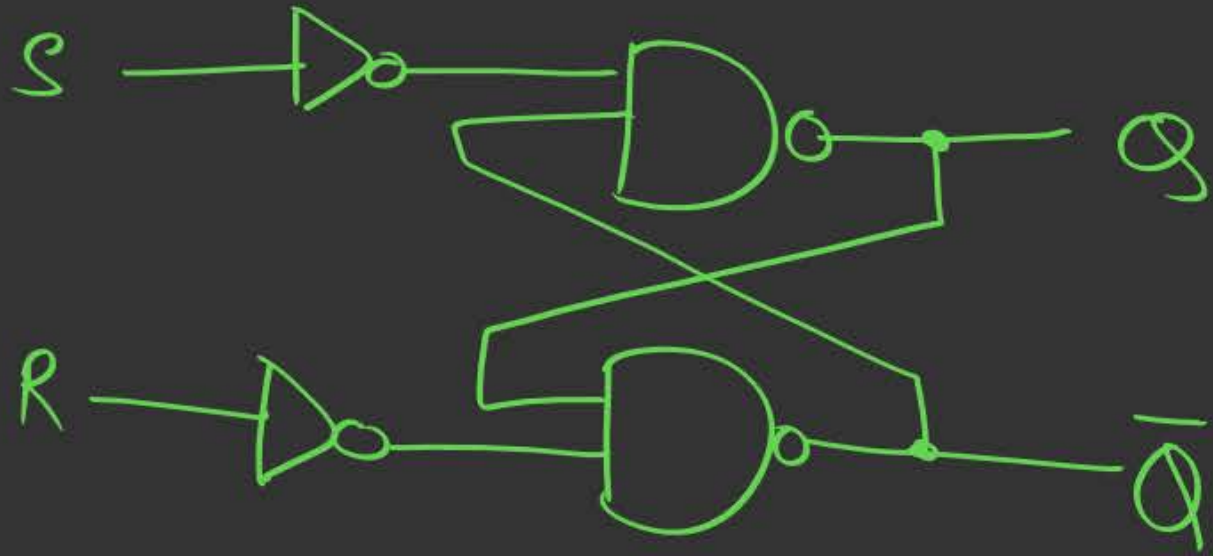


Telegram



Telegram Link for Aditya Jain sir: [https://t.me/AdityaSir PW](https://t.me/AdityaSir_PW)

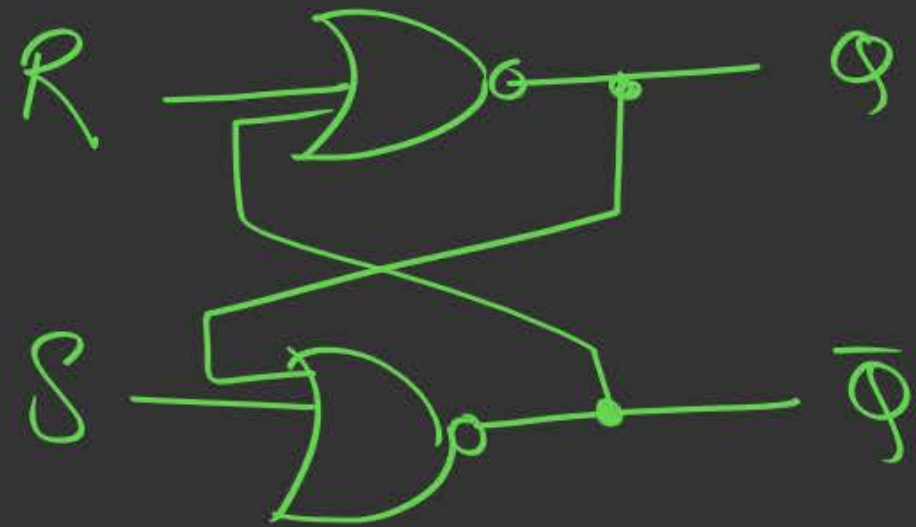
S-R Latch: (Set - Reset Latch)



S-R Latch using NAND

S	R	Q	$\bar{Q}$	
0	0	Q	$\bar{Q}$	→ Hold
0	1	0	1	→ Reset
1	0	1	0	→ Set
1	1	X	X	→ Invalid

S-R Latch using NOR



S	R	Q	$\bar{Q}$
0	0	Q	$\bar{Q}$
0	1	0	1
1	0	1	0
1	1	X	X

→ (Hold)

→ (Reset)

→ (Set)

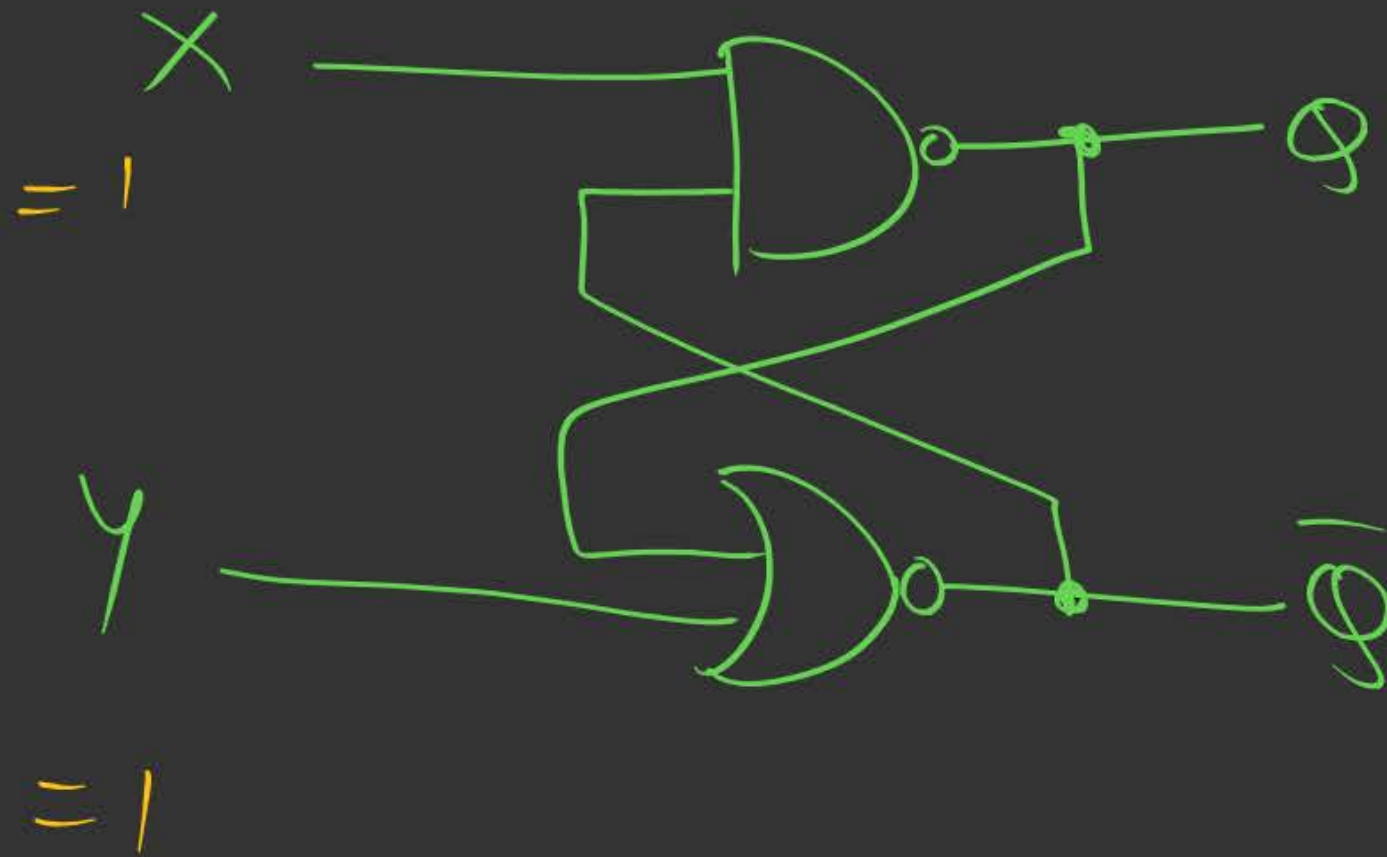
→ {invalid,
forbidden
Don't care}

NOR gate

X	Y	$\overline{X+Y}$
0	0	1
0	1	0
1	0	0
1	1	0

when any i/p is 1, the o/p is 0.

eg- Latch with NAND & NOR both.



X	Y	Q	$\bar{Q}$
0	0	1	0
0	1	1	0
1	0	0	1
1	1	1	0

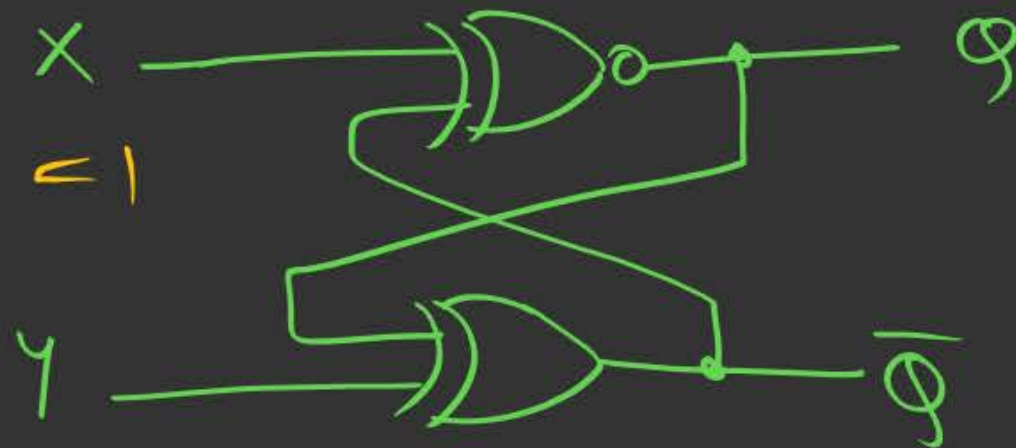
general obs:-

$$\overline{A} \oplus 0 = \overline{A}$$

$$A \oplus 0 = A$$

$$A \oplus 1 = \overline{A}$$

$$\overline{A} \oplus 1 = A$$



$$\begin{aligned} \overline{B} \odot 0 &= B \\ B \odot 0 &= \overline{B} \\ B \odot 1 &= B \\ \overline{B} \odot 1 &= \overline{B} \end{aligned}$$

X	Y	Q	$\overline{Q}$
0	0	Q	Q
0	1	Q	$\overline{Q}$
1	0	$\overline{Q}$	Q
1	1	$\overline{Q}$	$\overline{Q}$

→ (invalid)

→ (Hold)

→ (Toggle)

→ (invalid)

Flip Flop

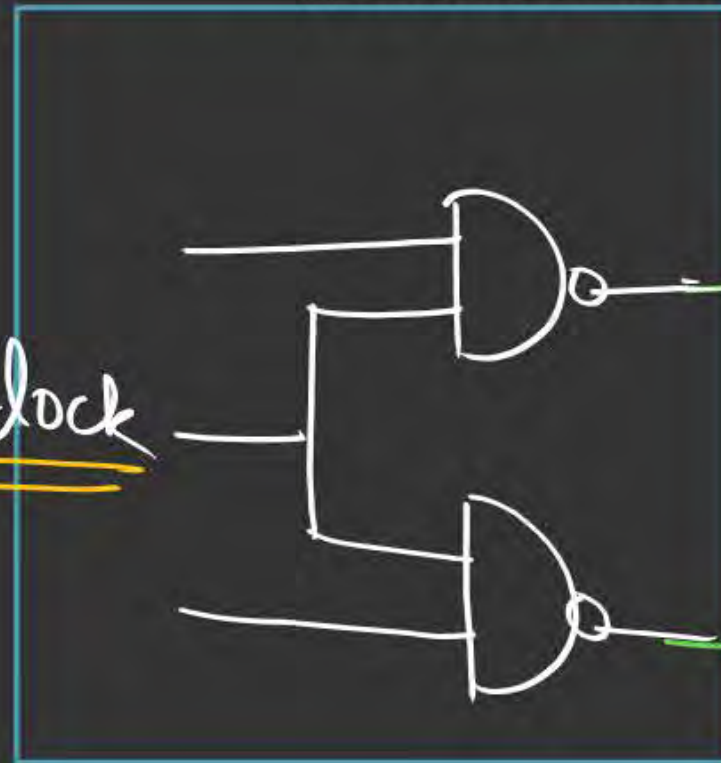


{ Control mechanism + Latch }

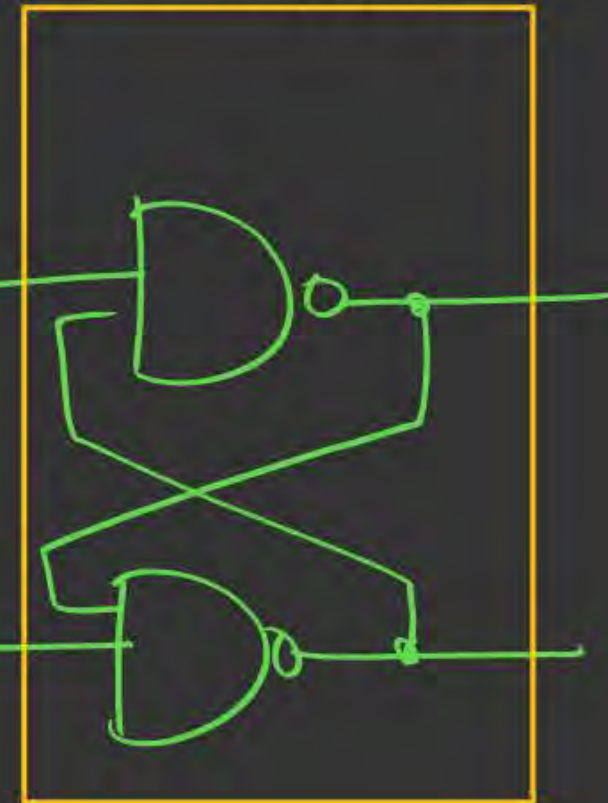
(Enable/Clock)

Enable / Clock

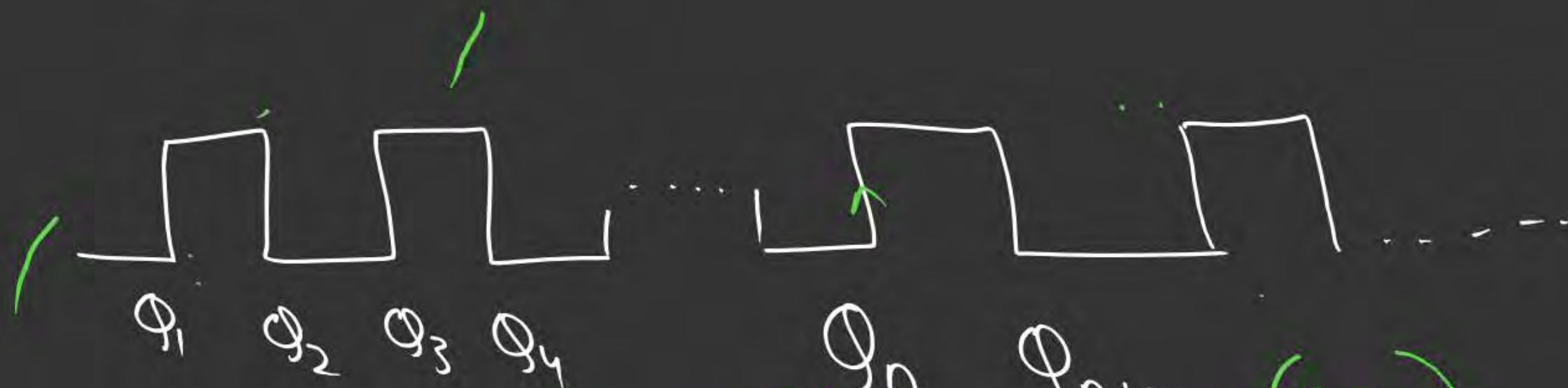
Control mechanism



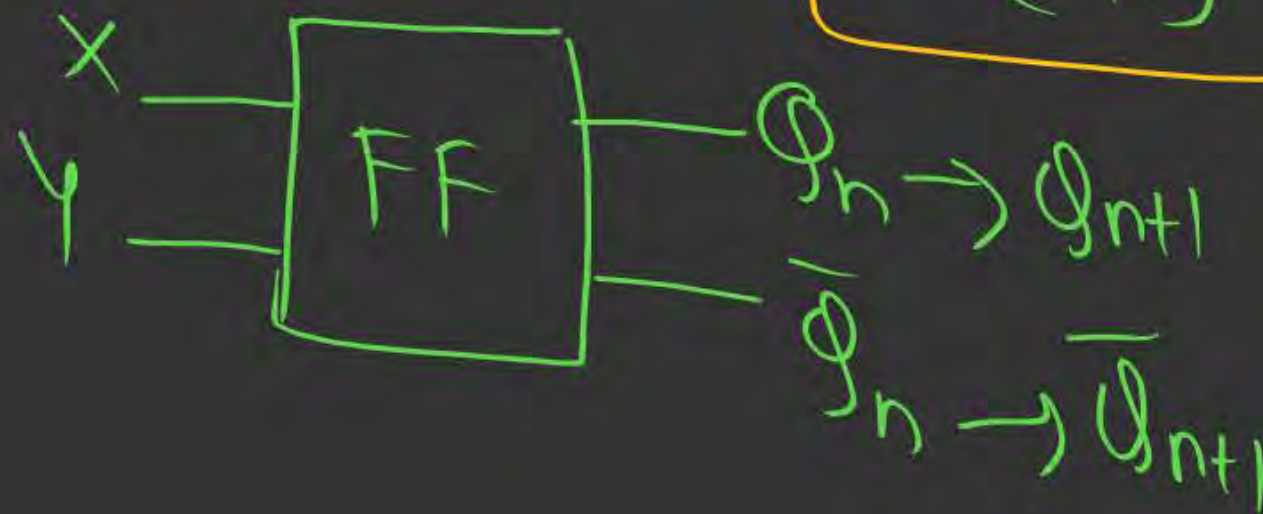
Latch



Clock



Q_n (prev) $\rightarrow$ n^{th} $Q_{n+1} \rightarrow$ (curr) $\downarrow$ $(n+1)^{\text{th}}$



FlipFlop:

★
✓ 1) Circuit Diagram

✓ 2) Truth Table

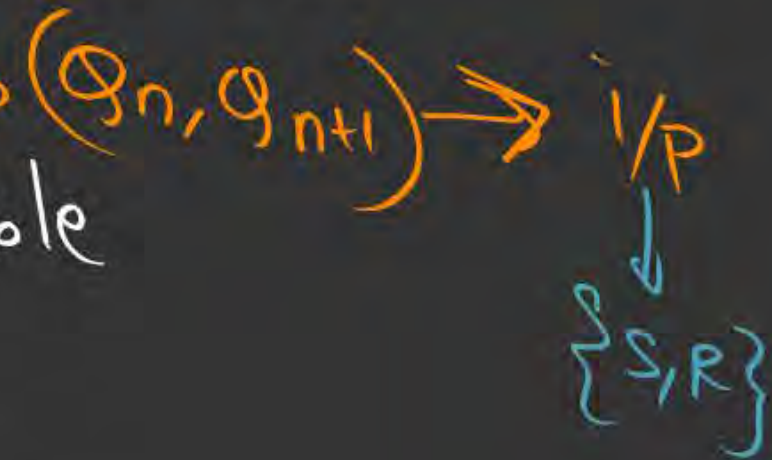
✓ 3) Characteristic table

✓ 4) Characteristic equation

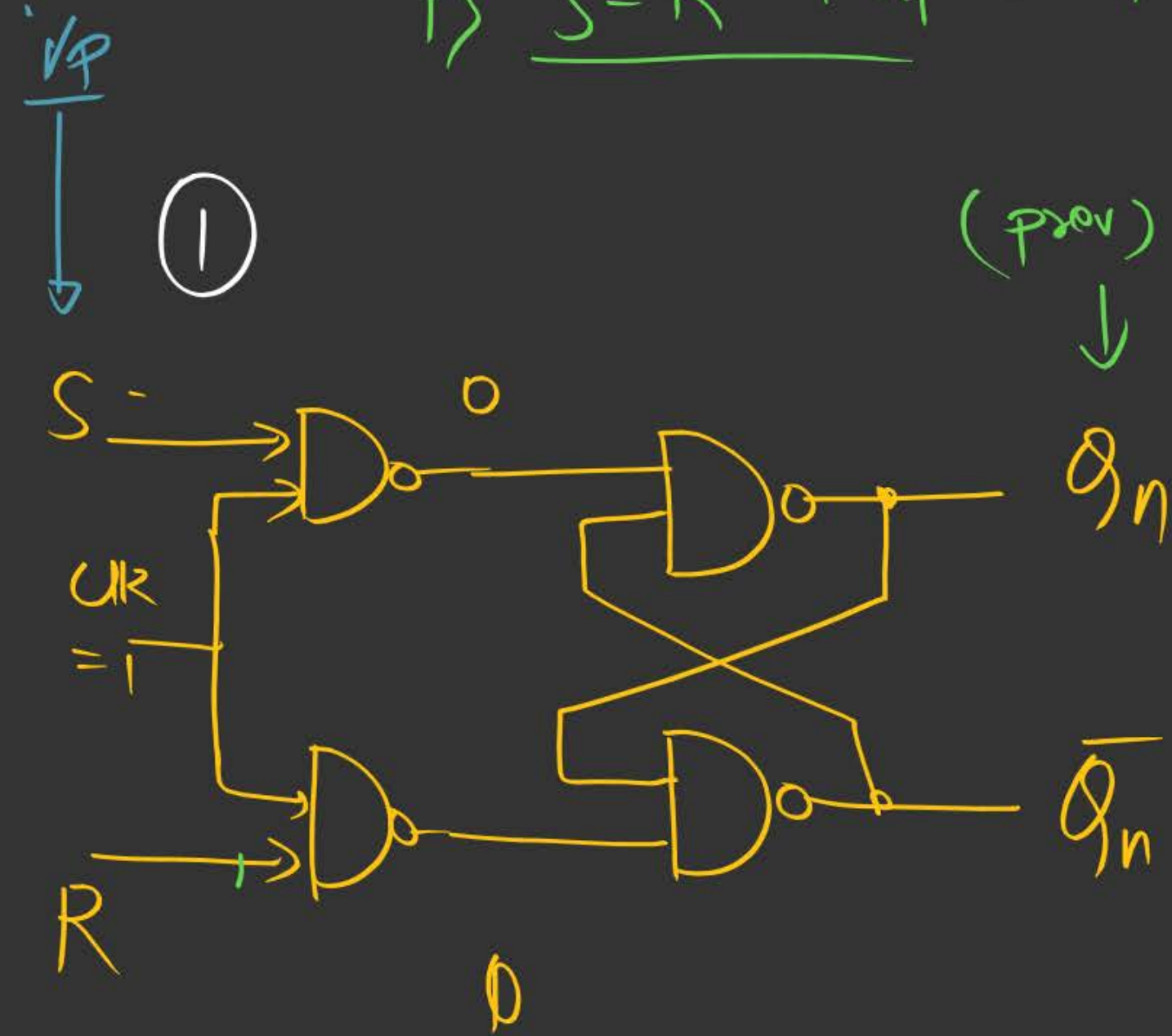
✓ 5) Excitation table

✓ 6) State Diagram.

$$Q_{n+1} \rightarrow (i/p + Q_n)$$



1) S-R Flip Flop (Set Rest FF)



(prev)
↓

② Truth Table

$Q_{n+1} \rightarrow$ Current O/P
 $Q_n \rightarrow$ prev O/P

S	R	Q_{n+1}	$\bar{Q}_{n+1}$
0	0	Q_n	$\bar{Q}_n$
0	1	0	1
1	0	1	0
1	1	X	X

{previous state}
memory

→ Hold

→ (Reset)

→ (Set)

→ (invalid)

FF active, $UR = 1$

3) Characteristic Table

→ present o/p.

$$\left\{ \begin{array}{l} \text{i/p: } S, R, Q_n \\ \text{o/p: } Q_{n+1} \end{array} \right\}$$

$$Q_{n+1}(S, R, Q_n)$$

S	R	Q_{n+1}	$\overline{Q_{n+1}}$
0	0	Q_n	$\overline{Q_n}$
0	1	0	1
1	0	1	0
1	1	X	X

S	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

3) Characteristic Table

$\left\{ \begin{array}{l} \text{IP: } S, R, Q_n \\ \text{O/P: } Q_{n+1} \end{array} \right\}$

$Q_{n+1}(S, R, Q_n)$

S	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1 ✓
0	1	0	0
0	1	1	0
1	0	0	1 ✓
1	0	1	1 ✓
1	1	0	X
1	1	1	X

0
1
2
3
4
5
6
7

→ dont care

④ Characteristic Equation

$$Q_{n+1}(S, R, Q_n) = \sum m(1, 4, 5) + \sum d(6, 7)$$

$$Q_{n+1} = S + \bar{R} Q_n$$

↓
dont care

(V.V. Imp)

Remember

$\bar{R}Q$

$S \backslash RQ$	$\bar{R}\bar{Q}$	$\bar{R}Q$	RQ	$R\bar{Q}$
$\bar{S}$	0	1	3	2
S	1	4	5	6

$\bar{R}Q$ (circled)
 S (circled)
 $\bar{R}\bar{Q}$ (circled)
 RQ (circled)

3) Characteristic Table $\rightarrow$ per row o/p.

$\left\{ \begin{array}{l} \text{i/p: } S, R, Q_n \\ \text{o/p: } Q_{n+1} \end{array} \right\}$

$Q_{n+1}(S, R, Q_n)$

5) Excitation table

(v.v. Imp)

S	R	Q_n	Q_{n+1}
0	0	0	0 ✓
0	0	1	1
0	1	0	0 ✓
0	1	1	0
1	0	0	1 ✓
1	0	1	1
1	1	0	X
1	1	1	X

4) $Q_{n+1} = S + \bar{R} Q_n$

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

⑤ Excitation table

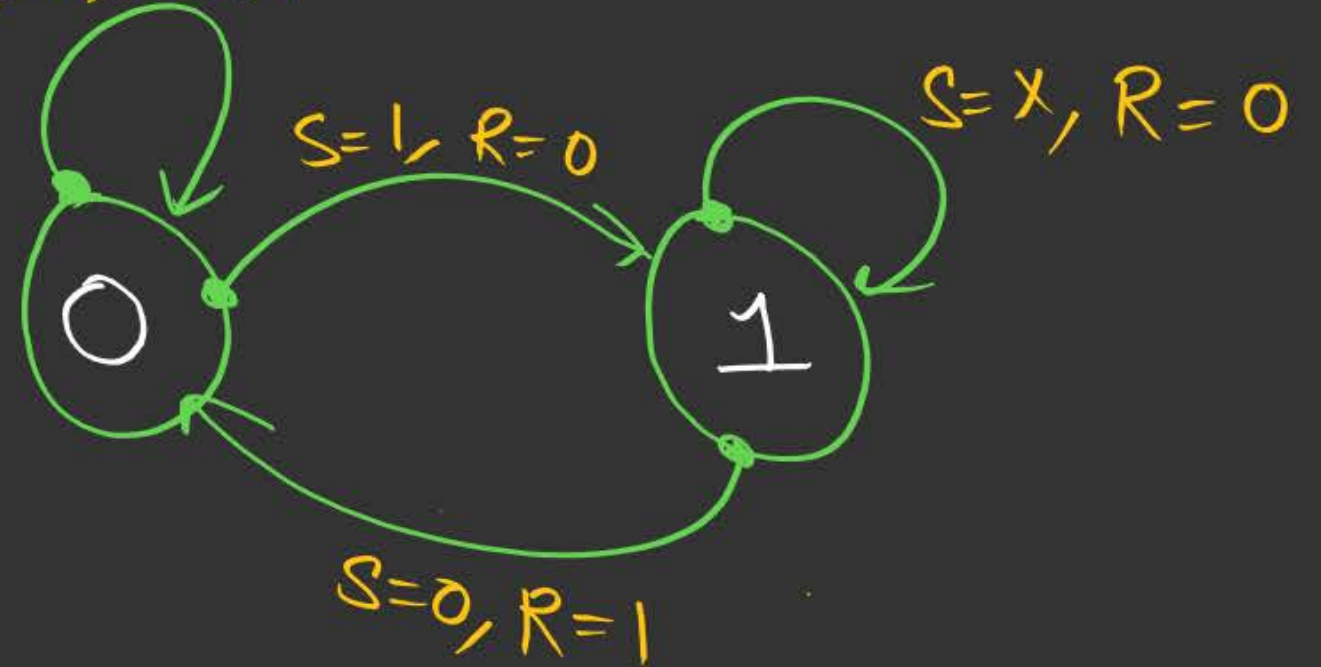
(v.v. Imp)

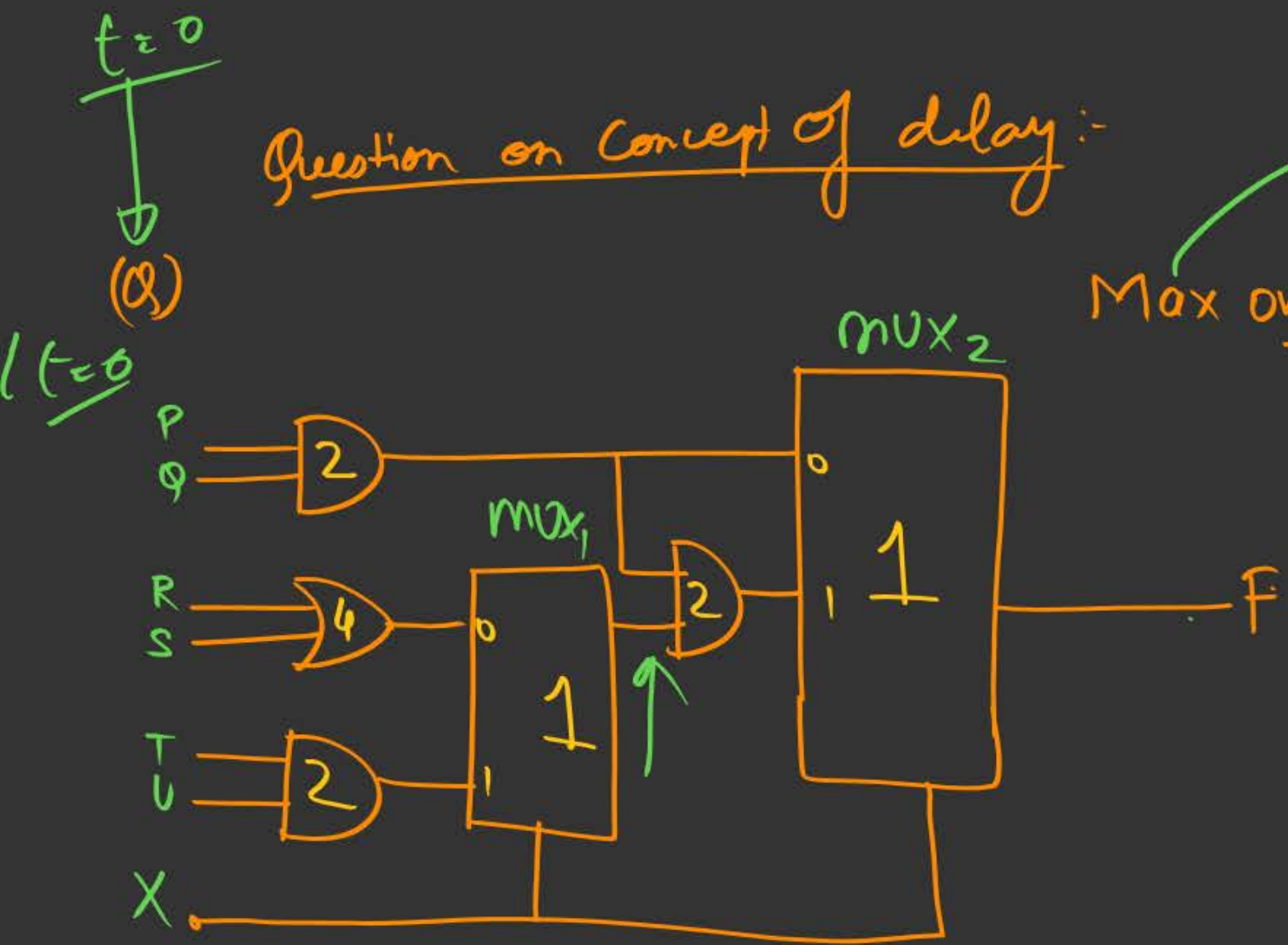
Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0



★ State Diagram

$S=0, R=X$





Max overall delay?

$$\max(3, 6) = \boxed{6}$$

$\downarrow$
ans

$$\underline{\underline{X=0}}$$

$$\text{delay} = T_{\text{and}} + T_{\text{mux}_2}$$

$$= 2 + 1 = \boxed{3}$$

$$\underline{\underline{X=1}}$$

$$\text{delay} = T_{\text{and}} + T_{\text{mux}_1} + T_{\text{and}} + T_{\text{mux}_2}$$

$$= 2 + 1 + 2 + 1 = \boxed{6}$$



2 mins Summary



Latches

S-R Flipflop

Delay Question



THANK - YOU

